



Department of Mechanical Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Mechanical Engineering

Course Code & Title: ME3493 Manufacturing Technology II

Name of the Faculty member (s): Mr.S.Maharajan, AP(Sr.Gr)/Mech

Innovative Practice Description

- **Unit / Topic:** Unit II / Capstan and Turret Lathe, Tumbler Gear Mechanism, Apron Mechanism and Half Nut Mechanisms
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4 & TLO5
- **Activity Chosen:** Theory to Practical
- **Justification:**
 - ✓ To understand the mechanisms and capstan and turret lathe operations by real time functioning of lathe.
 - ✓ To acquire the hands-on training in operating the lathe and know about the mechanisms involved in each operation in lathe.
 - ✓ To recollect the theoretical concepts of mechanisms by practical experience.
- **Time Allotted for the Activity:**

50 minutes for listening concepts and hands on experience in operating lathe by understanding the various mechanisms involved in all the parts in lathe.
- **Details of the Implementation:**

The students were asked to operate the mechanisms after listening the concepts behind the mechanisms of lathe. Student's skills were assessed by interacting with them by asking questions related to concepts and mechanisms of various parts of lathe.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO10	PO12	PSO2
CO2	3	3	3	1	1	1	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO4	PO9	PO10	PO12	PSO2
	3	2	3	3	3	3
Justification for correlation	Prerequisite knowledge in various mechanisms (ratchet and pawl mechanisms, gear transmissions) required to understand the real time applications.	To understand the complex operations performed in lathe by acquiring skills in real time practical experience.	By having elaborate discussion with fellow students for understanding the critical power transmission process. The operations were carried out in a group of students for better understanding	By asking questions to individuals and they gave clear explanation about mechanisms involved in lathe.	It is a continuous process of learning to enable the students to improve their practical skills with theory concepts	The students are provided with the manufacturing concepts, process plan and manufacturing solutions.

- **Images / Screenshot of the practice:**



Figure 1. Live Demo on Capstan and Turret Lathe operations

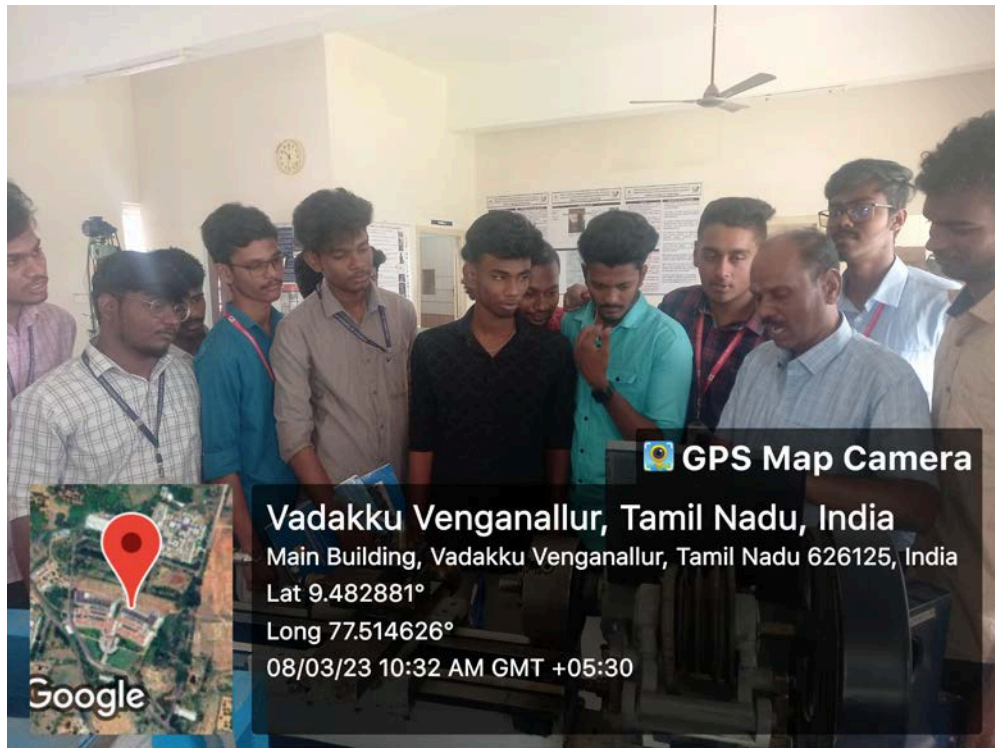


Figure 2. Live Demo on Lathe Thread Cutting Operation



Figure 3. Live Demo on Tumbler Gear Mechanism



Figure 4. Live demo on Lathe thread cutting procedure in Lathe



Figure 5. Live demo on Apron Mechanism in Lathe

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Concepts were clearly understood by the students. They improved the hands-on experience in operating various lathe by knowing the mechanisms behind all the parts of lathe.

- ❖ **Benefit of the practice:**

- ✓ By the hands-on training in Lathes, students can able to understand the complex mechanisms easily.
 - ✓ By interacting with fellow students, they improved the communication skills and team building activities.

- ❖ **Challenges faced in implementation:**

Short duration period for the training of Lathe machines is the huge challenge while the students are involving the real time practical session.

References:

- ❖ Hajra Choudhury, “Elements of Workshop Technology”, Vol.II., Media Promoters
- ❖ Rao.P.N “Manufacturing Technology – Metal Cutting and Machine Tools”, Tata McGraw-Hill, New Delhi, 2003.

Signature of Faculty Member

HOD

